

Sorbonne Université
Examen Deep Learning, Master MIND & MS2A
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Problem — Interpreting In-Context Learning

Recent analyses suggest that in-context learning (ICL) in Transformer models can be interpreted as an implicit gradient-based parameter update. In this exam we study a simplified mathematical form supporting this interpretation.

Notation:

- Vectors are column vectors.
 - Dot product: $a^T b$.
 - For matrices A, B with the same number of rows, $[A \ B]$ denotes column-wise concatenation.
 - For a vector x , we denote its r -th coordinate by $x[r]$. (Training examples are denoted x_i and are vectors.)
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Part A — Course (concepts)

- A1.** Explain (self-)attention as used in decoder modules of GPT (causal masking, Q/K/V, softmax, multi-head).
- A2.** Describe the elementary building block of a decoder layer.
- A3.** Explain what in-context learning (ICL) is in Transformer-based models.

Part B — Gradient Updates and Linear Attention

Consider a linear model

$$F(x) = Wx,$$

with $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, $x \in \mathbb{R}^{d_{\text{in}}}$ and $F(x) \in \mathbb{R}^{d_{\text{out}}}$.

Suppose we start from W_0 and train by steepest gradient descent on a training set $\{(x_i, y_i); x_i \in \mathbb{R}^{d_{\text{in}}}, y_i \in \mathbb{R}^{d_{\text{out}}}\}_{i=1}^N$.

Let $a_i = F(x_i) = Wx_i \in \mathbb{R}^{d_{\text{out}}}$ and $L_i = L(a_i, y_i)$ be the per-example loss. Assume for simplicity one gradient step per example, with learning rate $\gamma > 0$:

$$W \leftarrow W - \gamma \frac{\partial L_i}{\partial W}.$$

- B1. (Single example gradient).** For a single training example (x, y) , denote $a = F(x)$ and $L = L(a, y)$. Write the model componentwise as

$$a[j] = \sum_{\ell=1}^{d_{\text{in}}} W_{j\ell} x[\ell].$$

Show that

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial a} x^T.$$

(Indicate and check dimensions.)

- B2. (Accumulated update).** Let $W_0, \Delta W$ be the initialized parameter matrix and the update matrix respectively. Using the update rule above, show that after processing N examples the total update can be written

$$\Delta W = W - W_0 = \sum_{i=1}^N e_i x_i^T,$$

where each $e_i \in \mathbb{R}^{d_{\text{out}}}$ is an error-like term that can be expressed in terms of γ and $\frac{\partial L_i}{\partial a_i}$.

- B3.** Show that for any test input $x \in \mathbb{R}^{d_{\text{in}}}$,

$$F(x) = (W_0 + \Delta W)x = W_0 x + \sum_{i=1}^N e_i (x_i^T x).$$

- B4. (Linear attention definition).** Let $q \in \mathbb{R}^{d_{\text{in}}}$, $k_i \in \mathbb{R}^{d_{\text{in}}}$, $v_i \in \mathbb{R}^{d_{\text{out}}}$ be query, key, value column vectors. We stack keys and values as columns:

$$K = [k_1, \dots, k_N] \in \mathbb{R}^{d_{\text{in}} \times N}, \quad V = [v_1, \dots, v_N] \in \mathbb{R}^{d_{\text{out}} \times N}.$$

Define linear attention by

$$\text{Attn}(V, K, q) = \sum_{i=1}^N v_i (k_i^T q) = V(K^T q).$$

Rewrite the updated model $F(x)$ using this operator and identify precisely what are the values V , keys K , and query q in this model.

- B5.** Explain why this shows that a gradient-updated linear model can be viewed as attention over training examples. Hint: explain the roles of $x_i^T x$ and e_i .

Part C — Linearized Transformer Attention as an Implicit Update (ICL setting)

We consider ICL with a GPT-like decoder. The decoder is given demonstration examples (context) and a query example. For analysis, consider a simplified attention head without softmax:

$$\text{Attn}(V, K, q) = V K^T q.$$

Let Z be the matrix of token representations in the attention module, split as

$$Z = [Z_0 \ Z_q],$$

where:

- Z_0 are demonstration token representations,
- Z_q are query-side token representations (query text + answer prefix).

Keys, values and query are computed by linear projections:

$$K = W_K Z, \quad V = W_V Z, \quad q = W_Q x.$$

Clarification (tokens vs query vector). Z_0 contains the demonstration tokens, Z_q contains all the token vectors (columns) of the query. Attention is computed for *one token position* at a time: if x is the hidden state of the current token (one column of Z_q), then the attention query vector is $q = W_Q x$. If our objective is prediction, x could be the last token of Z_q .

Example (few-shot classification prompt).

Review: Great movie. Sentiment: Positive
 Review: Bad acting. Sentiment: Negative
 Review: Wonderful story. Sentiment: Positive
 Review: Boring film. Sentiment:

Tokens from the first three labeled reviews belong to Z_0 ; tokens from the last review and the prefix “Sentiment:” belong to Z_q . If we compute attention at the token “Sentiment:”, its hidden state is x (a column of Z_q) and the query vector is $q = W_Q x$.

C1. Let $F_{\text{ICL}}(q) = \text{Attn}(V, K, q)$. Show the decomposition

$$F_{\text{ICL}}(q) = W_V Z_q (W_K Z_q)^T q + W_V Z_0 (W_K Z_0)^T q.$$

C2. Define

$$W_{\text{base}} = W_V Z_q (W_K Z_q)^T.$$

Show that

$$F_{\text{ICL}}(q) = (W_{\text{base}} + \Delta W_{\text{ICL}})q,$$

and give an explicit expression for ΔW_{ICL} .

C3. (Interpretation and parallel with Part B). Interpret the decomposition of $F_{\text{ICL}}(q)$ obtained above. You can follow the steps below:

- (i) Explain the respective roles of $W_{\text{base}}q$ and $\Delta W_{\text{ICL}}q$.
- (ii) Show that

$$\Delta W_{\text{ICL}} = \sum_i v_i k_i^T$$

and identify precisely v_i, k_i (coming from demo tokens) and the query vector q .

- (iii) Both derivations in sections B and C lead to the same algebraic structure

$$\text{base term} + \sum_i v_i (k_i^T q).$$

State precisely what are the v_i, k_i, q for each case and the correspondence between the objects in both settings (training examples vs. demonstration tokens, gradients vs. values, inputs vs. keys).

C4. State two simplifications w.r.t. the generic self-attention block used to obtain this gradient-descent-like interpretation.